PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**(http://bmjopen.bmj.com/site/about/resources/checklist.pdf)** and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Soap versus sanitiser for preventing the transmission of acute respiratory infections in the community: a systematic review with meta-analysis and dose-response analysis
AUTHORS	Hoffmann, Tammy; Bakhit, Mina; Krzyzaniak, Natalia ; Del Mar, Chris; Scott, Anna Mae; Glasziou, Paul;

VERSION 1 – REVIEW

REVIEWER	Bajpai, Ram Keele University, School of Primary, Community, and Social Care
REVIEW RETURNED	02-Dec-2020

GENERAL COMMENTS	Introduction seems very short to understand what is known with reference to the given context and rationale for conducting this systematic review. Authors should add some more details. No information has been provided whether this review was registered with the PROSPERO or not. If it was not registered, please add your justification in the method section. As it is the secondary analysis of a previously published review using subset of articles, why this question wasn't answer previously in main review, and Authors mentioned that three authors independently reviewed the titles and abstracts of identified studies to assess eligibility for inclusion. Kindly add their initials. Did authors calculate agreement (such as kappa statistics) between reviewers? If so, please details in the method section. Authors should use Cochrane Collaboration's Risk of Bias (RoB) 2 instead of version 1. For I-squared statement, please add the range for various levels heterogeneities with appropriate citation. No justification was given for using random effects model. Authors must provide the appropriate reason with citation. A number of cluster RCTs were included in this report. However, nothing has mentioned in the analysis plan about clustering effect was adjusted in calculating effect estimate for a given cluster RCT. Please add the appropriate version of Review Manager. Discussion wasn't enough as it didn't provide comprehensive discussion about the research questions answered in this report.

REVIEWER	Khamis, Assem Hull York Medical School, Wolfson Palliative Care Research Centre

REVIEW RETURNED	07-Dec-2020

GENERAL COMMENTS	Peer Review: Soap versus sanitiser for preventing the transmission of acute respiratory infections: a systematic review with metaanalysis and dose-response analysis The article covers a current relevant public health topic which is the effectiveness of soap and sanitizers in protection against acute respiratory infections in community settings. The authors highlighted the importance of handwash within each subgroup of interventions and were able to present a dose-response analysis of effectiveness. The study emphasizes the importance of taking public health measures against the spread of infections based on available evidence in a community setting. Major revisions: • In the title, the authors should mention the community setting as the healthcare setting was excluded. Search dates • There might be quite a lot of evidence were published since March 2020 (start of the pandemic) for such a hot topic in the current times. I might suggest updating the search to be up-to-date evidence. Data analysis: • I think Figure A1 should be the primary/main analysis of this review, then the 2nd analysis whereas the handwashes were estimated. Another question, is there a possibility to subgroup analyze the studies based on follow-up durations, population (adults, children), and infections investigated? • Soap analysis – figure 3 1- Duration of intervention: Ram 2015 & Simmerman 2011 are both within a short period (<1 month) in influenza and ILI circumstances which are outbreak-based events compared to Nicholson 2014 which is almost a year. 2- Nature of infection and seasonality: Ram 2015 & Simmerman 2011 are both in influenza and ILI circumstances which are outbreak based events compared Nicholson 2014 which is general practice. Merging both types of studies regardless of the infection circumstances might send a confusing message while it could be that soap is effective in the long term and not in outbreak respiratory infections. • Hand sanitizer – figure 3 1- Infections: I would suggest adding a column to table 1 that specifies what infections are discussed in each study and make a separate table for the calculations. 2- Duration of intervention: it would be great if the authors highlighted the variability of interventions' duration as might be affected by the adherence to these interventions. • It would be great if the authors could highlight and summarize what are the control measures in each study and if there are any differences between studies.

	• It would be helpful if the authors could add a meta-analysis figure for the head-to-head trials of hand hygiene with hand sanitiser vs with soap and water.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1

 Introduction seems very short to understand what is known with reference to the given context and rationale for conducting this systematic review. Authors should add some more details.

Author response:

We have added additional information into the Introduction (Pg 4).

 No information has been provided whether this review was registered with the PROSPERO or not. If it was not registered, please add your justification in the method section.

Author response:

We have added the following to Methods:

Page 4, “This review was not registered on PROSPERO as this review is an additional analysis of a subset of trials from a published Cochrane systematic review.”

 As it is the secondary analysis of a previously published review using subset of articles, why this question wasn’t answer previously in main review, and Authors mentioned that three authors independently reviewed the titles and abstracts of identified studies to assess eligibility for inclusion. Kindly add their initials.

Author response:

The relevant authors initials have been included in the manuscript (pg. 5). The main Cochrane review was a review of all physical interventions to prevent ARI transmission and is a very big review, with a large number (67) of included trials. As such, it was not possible within that review to do the additional analyses (such as dose-response analysis) of individual categories of interventions (such as hand hygiene) - hence this review. This has been added into the paper (pg 4).

 Did authors calculate agreement (such as kappa statistics) between reviewers? If so, please details in the method section.

Author response:

It is not typical to calculate agreement between screeners when undertaking a systematic review. Rather, we discussed and resolved any disagreements to achieve consensus or with a third author. A clarifying statement has been included in the Methods (pg 5/6).

 Authors should use Cochrane Collaboration’s Risk of Bias (RoB) 2 instead of version 1.

Author response:

As mentioned earlier, the structure of this review was taken from the parent Cochrane review which was not required to use RoB2 as it was an update. To minimise confusion amongst readers who examine both articles, the authors preferred to maintain consistency and use the same RoB tool in both reviews.

 For I-squared statement, please add the range for various levels heterogeneities with appropriate citation.

Author response:

Thresholds for the interpretation of the I2 statistic may be misleading, as the importance of inconsistency depends on several factors. However, in accordance with the guidelines provided in the Cochrane handbook, we have added an outline for I2 interpretation in the Methods as follows:

Page 7, para 1: “We referred to the Cochrane Handbook in the interpretation of the I2 statistic, and were guided by the following ranges: 0% to 40%: may represent non-significant heterogeneity; 30% to 60%: may represent moderate heterogeneity; 50% to 90%: may represent substantial heterogeneity; 75% to 100%: considerable heterogeneity.”

 No justification was given for using random effects model. Authors must provide the appropriate reason with citation.

Author response:

A justifying statement has been added to the Methods as follows:

Page 7, para 1: “A random effects model was used because we expected some heterogeneity in the populations, interventions and outcomes of the included studies.”

 A number of cluster RCTs were included in this report. However, nothing has mentioned in the analysis plan about clustering effect was adjusted in calculating effect estimate for a given cluster RCT.

Author response:

A clarifying statement has been included in the Methods as follows:

Page 7, para 1: “To ensure we accounted for the clustering effect in the cluster-RCTs we extracted the appropriate cluster-adjusted estimates from the trial reports and used the generic inverse variance method of meta-analysis.”

 Please add the appropriate version of Review Manager.

Author response:

The most updated version of the Review Manager software has been cited in the manuscript.

 Discussion wasn't enough as it didn't provide comprehensive discussion about the research questions answered in this report.

Author response: see response above.

Reviewer: 2

 In the title, the authors should mention the community setting as the healthcare setting was excluded.

Author response:

The title has been amended to include reference to the community setting. It now reads as follows:

Page 1, "Soap versus sanitiser for preventing the transmission of acute respiratory infections in the community: a systematic review with meta-analysis and dose-response analysis"

 There might be quite a lot of evidence were published since March 2020 (start of the pandemic) for such a hot topic in the current times. I might suggest updating the search to be up-to-date evidence.

Author response:

Thank you for your comment. We are conscious of the apparent time lapse since March 2020, however we would like to note that the article was submitted to the BMJ Open for review in October 2020 and has been under review since. However, in March 2021, we conducted a search of clinical trial registries and not identified any completed RCTs of these interventions in the community since the outbreak of the pandemic. We are aware of 2 ongoing trials, however these are not yet completed and so can't be included in our analysis at this time.

 I think Figure A1 should be the primary/main analysis of this review, then the 2nd analysis whereas the handwashes were estimated. Another question, is there a possibility to subgroup analyse the studies based on follow-up durations, population (adults, children), and infections investigated?

Author response: We have made this change. The original Figure A1 is now Figure 3a.

As for subgroup analysis, while such differences may occur by setting and population subgroups, the trials are too sparse to consider each a special case. We agree that more research by setting and duration is warranted, particularly as these may effect adherence.

 Soap analysis – figure 3

1- Duration of intervention: Ram 2015 & Simmerman 2011 are both within a short period (<1 month) in influenza and ILI circumstances which are outbreak-based events compared to Nicholson 2014 which is almost a year.

2- Nature of infection and seasonality: Ram 2015 & Simmerman 2011 are both in influenza and ILI circumstances which are outbreak based events compared Nicholson 2014 which is general practice.

Merging both types of studies regardless of the infection circumstances might send a confusing message while it could be that soap is effective in the long term and not in outbreak respiratory infections.

Author response: See response to comment above about sub-group analysis.

 Infections: I would suggest adding a column to table 1 that specifies what infections are discussed in each study and make a separate table for the calculations.

Author response:

An additional column has been added to Table 1 with the type of infection studied. The authors would prefer to keep the calculations column adds valuable information to this table and would prefer to keep it within Table 1.

 Duration of intervention: it would be great if the authors highlighted the variability of interventions' duration as might be affected by the adherence to these interventions.

Author response:

A clarifying statement has been included in the Discussion (pg 12):

"The variable duration of the interventions (which ranged from 2 to 26 months) in the included studies may have impacted the reported intervention adherence and hence the comparability of it."

 It would be great if the authors could highlight and summarize what are the control measures in each study and if there are any differences between studies.

Author response:

An additional table containing a side-by-side summary of the interventions and controls in each study has been included as an online supplementary table (online supplementary file 2) and a summary of the controls added into the Results (pg 7):

“The majority of trials used existing handwashing practices as the comparator condition. One trial additionally provided education on hand hygiene, four provided health promotion education and provided participants with education on the prevention and treatment of respiratory tract infections. None of the studies provided control patients with any hand hygiene materials.”

 It would be helpful if the authors could add a meta-analysis figure for the head-to-head trials of hand hygiene with hand sanitiser vs with soap and water.

Author response:

Thank you for this suggestion. Only 3 of the included trials (Pickering 2013, Azor-Martinez 2018 and Savolainen-Kopra 2012) included both a soap arm and a sanitiser arm, which would enable this comparison. However, Savolainen-Kopra 2012 reports ‘infection episodes’ as an outcome (which includes both respiratory and gastrointestinal infections together) and would be inappropriate to pool with the other two trials (which focus on respiratory outcomes). As such, we did not think a meta-analysis of two trials would be appropriate.

VERSION 2 – REVIEW

REVIEWER	Bajpai, Ram Keele University, School of Primary, Community, and Social Care
REVIEW RETURNED	08-Apr-2021

GENERAL COMMENTS	Authors have appropriately responded to my previous comments. I have no further comments.

REVIEWER	Khamis, Assem Hull York Medical School, Wolfson Palliative Care Research Centre
REVIEW RETURNED	24-Apr-2021

GENERAL COMMENTS	The authors addressed all the comments.